Tentukan $q(t)$ bila pada $t=0$, $q=10$, $q'=0$ dan sketsa grafik dengan geogebra

1. $q'' + 4q' + 3q = 75$

↓

Persamaan Karakteristik :

$m^2 + 4m + 3 = 0$

$(m+3)(m+1) = 0$

$m_1 = -3 \vee m_2 = -1$

$q_h(x) = C_1 e^{-3x} + C_2 e^{-x}$

$f(x) = 75$

$f'(x) = 0$

$f''(x) = 0$

$0 + 4(0) + 3q = 75$

$q = 25 \rightarrow V(x) = 25$

$q = q_h(x) + V(x)$

$= C_1 e^{-3x} + C_2 e^{-x} + 25$

$q' = -3C_1 e^{-3x} - C_2 e^{-x}$

$t=0, q=10$

$q(0) = 10$

$C_1 + C_2 + 25 = 10$

$C_1 + C_2 = -15 \ldots (1)$

$t=0, q'=0$

$q'(0) = 0$

$-3C_1 - C_2 = 0 \ldots (2)$

(1) dan (2)

$C_1 + C_2 = -15$

$-3C_1 - C_2 = 0 \quad +$

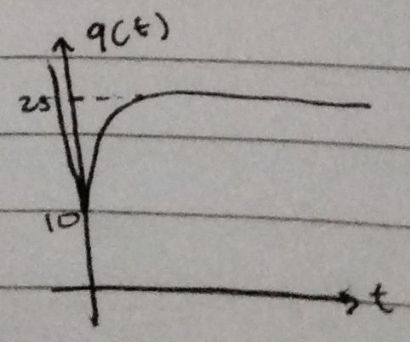
$-2C_1 = -15$

$\boxed{C_1 = 7{,}5}$

$7{,}5 + C_2 = -15$

$\boxed{C_2 = -22{,}5}$

$q = 7{,}5 e^{-3x} - 22{,}5 e^{-x} + 25$



No.

Date:

2. $q'' + 4q' + 4q = 75$

↓

Karakteristik Persamaan:

$m^2 + 4m + 4 = 0$ $\qquad q_h(x) = e^{-2x}(C_1 + C_2 x)$

$(m+2)(m+2) = 0$

$m_1 = -2 \vee m_2 = -2$

$f(x) = 75$

$f'(x) = 0$ $\Big\}$ $0 + 4(0) + 4q = 75$

$f''(x) = 0$ $\qquad q = 18{,}75 \rightarrow v(x) = 18{,}75$

$q = e^{-2x}(C_1 + C_2 x) + 18{,}75$ $\qquad q' = -2e^{-2x}(C_1 + C_2 x)$

$q(0) = 10$

$e^{-2(0)}(C_1 + C_2(0)) + 18{,}75 = 10$

$C_1 + 18{,}75 = 10$

$\boxed{C_1 = -8{,}75}$

$q' = u'v + uv' \quad \Big\{ u = e^{-2x} \quad v = C_1 + C_2 x$

$= -2e^{-2x}(C_1 + C_2 x) + e^{-2x} C_2 \quad \Big\} u' = -2e^{-2x} \quad v' = C_2$

$= -2C_1 e^{-2x} - 2C_2 x e^{-2x} + C_2 e^{-2x}$

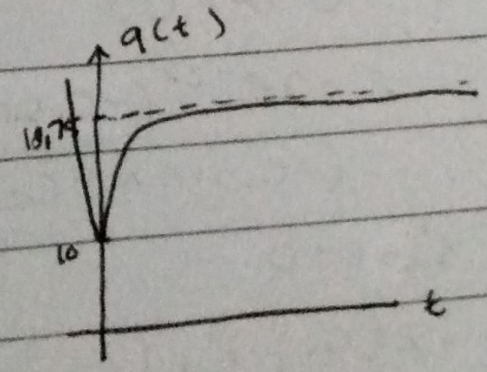
$q'(0) = 0$

$-2C_1 + C_2 = 0$

$-2(-8{,}75) + C_2 = 0$

$\boxed{C_2 = -17{,}5}$

$q = e^{-2x}(-8{,}75 - 17{,}5x) + 18{,}75$



3. $q'' + 4q' + 5q = 75$

↓

Persamaan Karakteristik :

$m^2 + 4m + 5 = 0$

$m_{1,2} = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-4 \pm \sqrt{16-4\cdot1\cdot5}}{2\cdot1}$

$= \frac{-4 \pm \sqrt{16-20}}{2} = \frac{-4 \pm \sqrt{-4}}{2}$

$= -2 \pm j$

$q_h(x) = e^{-2x}(C_1 \cos x + C_2 \sin x)$

$f(x) = 75$

$f'(x) = 0$

$f''(x) = 0$

$0 + 4(0) + 5q = 75$

$q = 15 \rightarrow V(x) = 15$

$q = e^{-2x}(C_1 \cos x + C_2 \sin x) + 15$

$q(0) = 10$

$e^{-2(0)}(C_1 \cos 0 + C_2 \sin 0) + 15 = 10$

$1(C_1) + 15 = 10$

$\boxed{C_1 = -5}$

$q' = u'v + uv'$

$= -2e^{-2x}(C_1 \cos x + C_2 \sin x) + e^{-2x}(-C_1 \sin x + C_2 \cos x)$

$u = e^{-2x}$

$u' = -2e^{-2x}$

$v = C_1 \cos x + C_2 \sin x$

$v' = -C_1 \sin x + C_2 \cos x$

$q'(0) = 0$

$-2(C_1) + 1(C_2) = 0$

$-2C_1 + C_2 = 0$

$-2(-5) + C_2 = 0$

$\boxed{C_2 = -10}$

$q = e^{-2x}(-5\cos x - 10 \sin x) + 15$

